



Lesson 2: Calculating Metrics in Microsoft Excel

Business Analytics Course | Spring 2024





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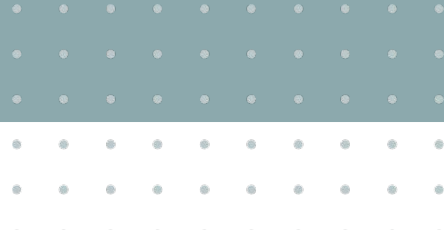


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01.

Review of Week 1 Learnings





Measure of Central Tendency

- **Mode:** The most frequently occurring value in your data set
- **Mean:** The average value in the data set
- **Median:** The middle value in the data set

Measure of Dispersion

- **Absolute Deviation:** The average distance of all of the elements of the elements in data set from the mean of the same data set
- **Variance:** A measurement of spread between numbers in a dataset
- **Standard Deviation:** The Square Root of variance
- **Quartiles:** Quartiles separate the data set in to quarters, or quartiles
- **Range:** The difference between the largest and the smallest observation in the data

Useful Excel Functions



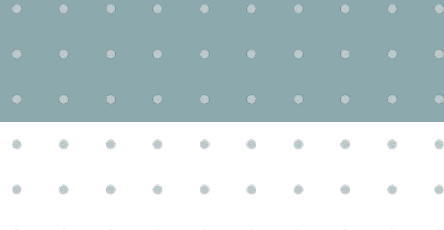
=AVERAGE()	Calculates the mean of the selected data
=MEDIAN()	Calculates the median of the selected data
=MODE()	Calculates the mode of the selected data
=AVERAGEIF()	Calculates the average of data IF it satisfies a condition
=STDEV()	Calculates the standard deviation of a SAMPLE
=VAR()	Calculates the variance of a SAMPLE

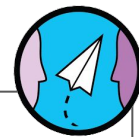




02.

Business Metrics Overview





What is Business Metrics?

Standardized quantitative measurements used to track, assess, and analyze specific data, performance, or conditions in various fields and industries.

Key Benefits of Tracking Business Metrics



Performance improvement: Tracking the right business metrics tells you how well or poorly the business is doing and provides direction for how to improve operations



Identifying problems: Analyzing business metrics can help identify emerging problems in time to correct them before they become major pain points.



Comparative analysis: Tracking business metrics reveals whether the business is over- or underperforming on key industry benchmarks.



Communication: Reporting business metrics is a vital communications tool for customers, shareholders, employees or society at large.

Business Metrics by Business Types



Retail



Sales Revenue
= $\text{SUM}(\text{Quantity Sold} \times \text{Price})$

Foot Traffic
= $\text{COUNT}(\text{Number of Visitors})$

Restaurant



Average Customer Bill
= $\frac{\text{SUM}(\text{Revenue per Day})}{\text{Number of Customers}}$

Table Turnover Rate
= $\frac{\text{Number of Meals Served}}{\text{Seating Capacity}}$

Hospitality



Occupancy Rate = $\frac{\text{Rooms Sold}}{\text{Total Rooms Available}}$

Average Daily Rate = $\frac{\text{Room Revenue}}{\text{Number of Rooms Rented}}$

Example



You sell homemade jams at a local farmer's market. In the month of May, you sold the following:



Strawberry Jam Jars:
Selling Price: \$5/jar
Cost: \$2/jar
Unit Sold: 100 jars



Blueberry Jam Jars:
Selling Price: \$6/jar
Cost: \$3/jar
Unit Sold: 60 jars



Peach Jam Jars:
Selling Price: \$4/jar
Cost: \$1/jar
Unit Sold: 40 jars

Calculate the following: Total Revenue, Total Cost of Goods Sold, Total Profit

Total Revenue = $(5 \times 100) + (6 \times 60) + (4 \times 40) = 500 + 360 + 160 = \$1,020$

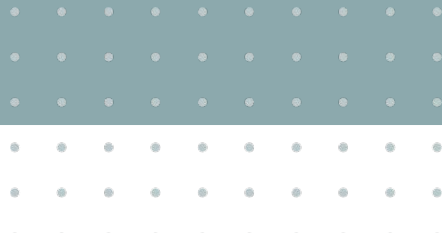
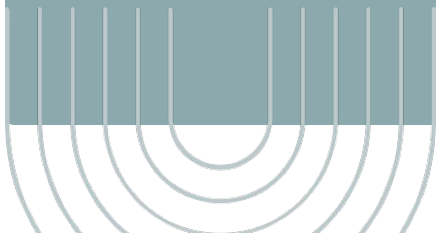
Total Cost of Goods Sold = $(2 \times 100) + (3 \times 60) + (1 \times 40) = 200 + 180 + 40 = \420

Total Profit = $\$1,020 - \$420 = \$600$



03.

Calculating Metrics In Excel



Common Excel Functions for Metrics – 1



COUNT Function

The **COUNT** function is a premade function in Excel, which counts cells with numbers in a range.

It is typed `=COUNT`

Note: The **COUNT** function only counts cells with numbers, **not** cells **with letters**. The COUNTA function is better used if the cells have letters.

Count practice: <https://excelexercises.com/practice.html?lesson=25>

SUM Function

The **SUM** function is a premade function in Excel, which adds numbers in a range.

It is typed `=SUM`

Sum practice: <https://excelexercises.com/practice.html?lesson=1>



Common Excel Functions for Metrics – 2



IF Function

The **IF** function is a premade function in Excel, which returns values based on a **true** or **false condition**.

It is typed `=IF` and has 3 parts:

```
=IF(logical_test, [value_if_true], [value_if_false])
```

The **condition** is referred to as `logical_test`, which can check things like:

- If a number is **greater than** another number `>`
- If a number is **smaller than** another number `<`
- If a number or text is **equal** to something `=`

IF practice: <https://excel-practice-online.com/functions/if-function/#practice>



Same Example Through Excel



You sell homemade jams at a local farmer's market. In the month of May, you sold the following:

B	C	D	E
	Strawberry Jam Jars	Blueberry Jam Jars	Peach Jam Jars
Selling Price:	\$5	\$6	\$4
Cost:	\$2	\$3	\$1
Unit Sold:	100	60	40

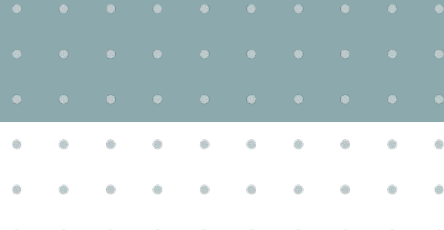
Calculate the following: Total Revenue, Total Cost of Goods Sold, Total Profit

Total Revenue	\$ 1,020.00	=SUMPRODUCT(C5:E5,C7:E7)
Total Cost of Goods Sold	\$ 420.00	=SUMPRODUCT(C6:E6,C7:E7)
Total Profit	\$ 600.00	=C10-C11



04.

Assignment



Excel Assignment



You work for a snack food distribution company that sells products to grocery stores across the East and West regions. The data table shows sales transactions from January for various product categories. Please download the sheet below (File → Download → Microsoft Excel) and complete the assignment :)

To check the answer, please see the following slides

ID	Date	Region	City	Category	Product	Qty Sold	UnitPrice	Total Sales
ID07351	1-Jan	East	Boston	Bars	Carrot	33	1.77	95.58
ID07352	4-Jan	East	Boston	Crackers	Whole Wh	87	3.49	520.01
ID07353	7-Jan	West	Los Angele	Cookies	Chocolate	58	1.87	90.27
ID07354	10-Jan	East	New York	Cookies	Chocolate	82	1.87	177
ID07355	13-Jan	East	Boston	Cookies	Arrowroot	38	2.18	37.8
ID07356	16-Jan	East	Boston	Bars	Carrot	54	1.77	78.48
ID07357	19-Jan	East	Boston	Crackers	Whole Wh	149	3.49	57.97
ID07358	22-Jan	West	Los Angele	Bars	Carrot	51	1.77	97.72
ID07359	25-Jan	East	New York	Bars	Carrot	100	1.77	77.88
ID07360	28-Jan	East	New York	Snacks	Potato Chi	28	1.35	40.71
ID07361	31-Jan	East	Boston	Cookies	Arrowroot	36	2.18	36.45

Assignment Link: [Business Metric Assignment](#)

Excel Assignment Answer



1. What was the total quantity sold across all products?

Total Quantity Sold
716
=SUM(G4:G14)

5. What is the average quantity purchased per transaction?

Average Quantity purchased
65.09
=AVERAGE(G4:G14)

2. What was the total sales for the East Region? West Region? (Hint: Use **SUMIF**)

East Revenue
\$1,121.88
=SUMIF(C4:C14,"East",I4:I14)

West Revenue
\$187.99
=SUMIF(C4:C14,"West",I4:I14)

6. What is the maximum unit price for an item?

Maximum Unit Price
\$3.49
=MAX(H4:H14)

3. How many transactions occurred in Los Angeles? (Hint: Use **COUNTIF**)

Los Angeles Transaction
2
=COUNTIF(D4:D14,"Los Angeles")

7. What is the range of total purchase amounts?

Range of Total Sales
\$483.56
=max(I4:I14)-min(I4:I14)

4. What percentage of total sales came from the East region?

% Sales from East
85.65%
=SUMIF(C4:C14,"East",I4:I14)/SUM(I4:I14)



THANK YOU

Have any question?